

Sanjana Srivastava

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EDUCATION

University of Virginia

Master of Science in Data Science

Current GPA: 4.00

May 2025

Indian Institute of Technology, Roorkee

Master & Bachelor of Technology in Geological Technology, Mathematics

CGPA: 8.202

July 2022

SKILLS

Languages: HTML, CSS, Python, R, SQL, C/C++, Java, JavaScript, Kotlin, $\text{\LaTeX}$

Packages: MongoDB, MySQL, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Git, AWS, GCP, Azure, Tableau, Hadoop, Spark, RAG, Docker, Linux

EXPERIENCE

AI Engineer, Deloitte (Capstone Project)

Oct 2024 – May 2025

- Built LLM evaluation pipeline using NLP and LLM-as-a-judge (GPT, Claude) for Deloitte's multi-domain clients
- Benchmarked DeepEval, MLFlow, Bedrock, TruLens, and RAGAS across 10+ datasets in a holistic [leaderboard](#)
- Identified and evaluated 8 key metrics including summarization, accuracy, hallucination, and toxicity detection

AI Research Scientist, University of Virginia, VA

June 2022 – May 2024

Advisors - Dr. Sana Syed, Dr. Donald E. Brown

- Developed Machine Learning models for disease diagnosis using tissue images, clinical, and transcriptomic data
- Collaborated with doctors to apply ML techniques for gut function analysis and prescriptive modeling
- Published and presented at top research conferences Nature, ESPGHAN, DDW, NASPGHAN, and ASTMH

Software Engineer Intern, BNY Mellon

Summer 2021

- Built a production-grade XML-to-CSV parser in Java, enabling data transformation for financial systems
- Implemented A/B testing for UI workflows and proposed performance optimization across internal dashboards

Software Engineer Intern, ZestMoney

Summer 2019

- Constructed a Java payment gateway, reducing transaction latency & increasing conversion rate by 5%
- Designed Android UI using MVVM architecture and Retrofit, accelerating user signups by 30% and enhancing UX

Content Writer, Rep3

Spring 2022

- Constructed a Java payment gateway, reducing transaction latency & increasing conversion rate by 5%
- Designed Android UI using MVVM architecture and Retrofit, accelerating user signups by 30% and enhancing UX

PROJECTS

Evaluating efficacy of synthetic images generated using diffusion models, [\[Code\]](#)

- Developed diffusion models to generate histology images patches conditioned on 6 nuclei locations
- Validated synthetic patches by improving downstream U-Net segmentation with a 5.2% gain in Dice score

Deep learning understanding of diseases using medical imaging and video data, [\[Code\]](#)

- Implemented supervised and unsupervised image disease detection using Neural Networks and Computer Vision
- Performed Gaussian clustering methods to identify recurring visual patterns in diseased biopsies

Text-driven latent diffusion model for localized fashion image editing, [\[Code\]](#)

- Constructed a U-Net-based conditional diffusion model for image editing guided by natural language text prompts
- Built a CLIP-guided segmentation model to localize garment regions from natural language prompts

Humorous image captioning system, [\[Code\]](#)

- Designed a self-attentive encoder-decoder framework using CLIP to generate humorous captions for images
- Devised an image retrieval model using RAG and prompt engineering

Correlating disease gene signature with imaging data, [\[Code\]](#)

- Designed a deep learning framework to identify image features correlated with functional gene clusters
- Identified important gene signatures and their correlation with visual patterns in biopsies

Buy and Sell application, [\[Code\]](#)

- Developed an intranet application for buying, selling and requesting goods among 8,000 campus residents
- Features include categorization of goods, subscribing, filtering, and searching based on users input

Petrographic characterization of a chondrite sample | Advisor - [Dr. Nachiketa Rai](#), IIT Roorkee

- Investigated mineralogy present in chondrite section leveraging Decision Tress and Linear Regression
- Performed data mining and data analysis on back-scattered electron images for predictive modeling

PUBLICATIONS & TALKS

- **Efficacy Of Synthetic Histopathological Images In Enhancing Nuclei Segmentation Tasks**
[S. Srivastava](#), A. Shrivastava, S. Rhoads, P.T. Fletcher, S. Syed, D.E. Brown. [Under Review]
- **What Is Normal? Characterization Of Control Pediatric Duodenal Biopsies Using Clinical Data, Machine Learning Image Analysis, And Transcriptomics**
F. Rhoads, J. Sessions, [S. Srivastava](#), F. Zulqarnain, V. Jain, ..., S. Syed. [Poster Presentation] *ESPGHAN 2024*
- **Machine-learning-based integrative-‘omics analyses reveal immunologic and metabolic dysregulation in environmental enteric dysfunction**
F. Zulqarnain, X. Zhao, K. Setchell, Y. Sharna, P. Fernandes, [S. Srivastava](#), ..., S. Syed. [\[Link\]](#) *iScience 2024*
- **Identifying Metabolic Signatures Of Environmental Enteric Dysfunction In Pakistani Mothers And Children Using Tissue-Specific Metabolic Modeling**
S. Syed, F. Zulqarnain, [S. Srivastava](#), W. Khan, I. Nisar. [Poster Presentation] *Bill & Melinda Gates Foundation Grand Challenges Annual Meeting 2023*
- **Quantitative Morphometry and Machine Learning Model to Explore Duodenal and Rectal Mucosal Tissue of Children with Environmental Enteric Dysfunction**
M. Khan, Z. Jamil, L. Ehsan, F. Zulqarnain, [S. Srivastava](#), ..., S. Syed. [\[Link\]](#) *ASTMH’ 2023*

AWARDS & ACHIEVEMENTS

- Recipient of **IIT Roorkee Graduation Award: Nayyar Award for Excellence in Communication** given to 1 student with exceptional communication skills from graduating batch.
- Recipient of **Indian Ministry of Human Resource and Development (MHRD) Assistantship** for meeting graduate CGPA criterion, demonstrating academic excellence exceeding required limit.
- Cleared **Joint Entrance Examination (JEE) with 99.6 percentile**.

POSITIONS OF RESPONSIBILITY & EXTRA CURRICULARS

- **Teaching Assistant** for a graduate level statistical learning course at University of Virginia
- **Project Leader** at Information Management Group - Official Coding Society of IIT Roorkee
- **Editor-in-chief** at Geek Gazette - Official Technical Magazine of IIT Roorkee
- **Actor/Director/Producer** at Dramatics Section, IIT Roorkee
- **Council Member** of Hostel Management at Kasturba Bhawan, IIT Roorkee
- **Team Member** of Table Tennis National Sports Organization, IIT Roorkee